

Chatbot as a persuasive technology to promote responsible recycling in the city of Lima

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Abstract—Persuasive technologies have gained popularity around the world. This type of technology aims to change the behavior of people in different circumstances using the persuasion method. Within this classification of technologies we have virtual assistants whose function is to interact in a personalized way with the user. This research aims to provide a solution to the low levels of recycling habits in citizens, which is caused by inefficient programs and/or activities applied by authorities, educational centers and households of socioeconomic levels. low fees by promoting recycling issues. We proposed the use of a virtual assistant to increase knowledge and show the benefits of recycling, in order to generate an improvement in the behavior of the population. To validate this research, two knowledge tests on recycling issues were administered to 69 university students at the beginning and end of the process. After using our virtual assistant, the participants appeared and increased their knowledge by 70.49% .

Index Terms—Recycle, chatbot, habits, artificial intelligence.

I. INTRODUCTION

Currently, topics on recycling have been heard in various sources. However, the lack of recycling habits among citizens continues to cause environmental pollution. Since the increase in solid waste management campaigns, different institutions have implemented infrastructure so that citizens can recycle, but this has not been enough for them to implement said activity. Generating recycling habits is difficult, since the number of universities that have institutional environmental plans or programs are few [1], in the same way in households with low socioeconomic levels, education is not encouraged and environmental culture [2]. Likewise, the programs that exist to inform the population are repetitive and traditional, in addition to this there is no follow-up to ensure that what is learned in the campaigns and/or programs is applied [3].

For this reason, all existing persuasive technologies were investigated, which can influence the change in the behavior of people in relation to recycling. Various technologies were found in our investigation. Among all of them, the chatbot was chosen, since it met the criteria that a technological solution should have to change habits [4]. With the chosen technology, it is intended to resolve doubts or queries about solid waste management in real time in order to motivate recycling habits in Peruvians. For the development of the project, the support of 69 university students was requested, who participated in a validation process of our solution. Finally, the results presented in our study have a confidence level of 90% and a margin of error of 10%. Next, the structure of our investigation will be described. In section two, the theoretical framework will be carried out, where we will detail the main concepts used. In section three the main contribution is developed, which will explain the selection of the chatbot, its implementation and the experiment to see the efficiency of the solution. In section four, we will specify the results obtained from the previous section. Finally, we will mention the conclusions obtained in relation to the results of the experiment.

II. THEORETICAL FRAMEWORK

A. Solid Waste Management

- Solid waste: Are all those substances, products or derivatives that are in a solid state or decomposition of which your generator has availability, or has the obligation to take it with you, by virtue of what is stated in national standards or of the risks they cause to health and the environment. This definition also includes all waste that is generated by natural events [5].
- Recycling: Recycling is necessary worldwide in order to improve the quality and state of the environment,

as well as protect the health of people and animals. The recycling of paper, plastic, glass, metals and other materials, makes it easier for us to save essential materials that are extracted from nature and increase the useful life of sanitary landfills in our cities, where every day observes that there is less space to dispose of waste [6].

B. Persuasive Technology

Persuasive technology is an area of research focused on technologies, concepts, design principles, theories, evaluation, strategies, etc. that produce a change in the attitudes or behaviors of users through the use of technologies such as virtual assistants, serious games, augmented reality, social networks, etc.

C. Chatbot

They are software created under the digital umbrella of artificial intelligence and serve to automate processes that are executed without the need for human intervention [7].

The Chatbot is a mechanism that has been developed according to artificial intelligence, machine learning and natural language processing, chatbots have grown extraordinarily so that it is currently possible to distinguish them according to the service they offer, these are classified as:

- Sales-oriented Chatbot: these are designed to make sales of products or services of the different organizations that have decided to use these assistants [8].
- ChatBot oriented to customer service, they have been developed to be able to solve the doubts that customers have regarding a certain service [8].
- ChatBot oriented to news and content, these are implemented through instant messaging channels, their main orientation is the sending of massive content through these different channels [8].

In the same way, the development of these systems has caused the generation of different applications today, daily the world is immersed in their use thanks to systems such as Siri, Google Assistant, etc [8].

D. Cloud provider

There are various cloud providers that offer the service of assistant for the creation of chatbots, for which a comparison is made to determine which of the providers is suitable for the development of our virtual assistant.

E. Service Cloud

- IBM Watson Assistant: It is a system based on Artificial Intelligence (AI), which is based on the most recent innovations in machine learning. Watson is an open, multi-cloud platform that enables any user to automate the life cycle of artificial intelligence [9].
- App Services AZURE: It is a tool that allows you to create and host web applications, mobile back-ends and restful APIs in the programming language of your choice without managing the infrastructure [10].

III. STATE OF ART

Several investigations have been carried out on remote supervision for online exams. An overview of these is presented below.

In [11] they propose to demonstrate that the therapeutic results under the use of mobile applications such as the chatbot are as effective as its face-to-face modality. The chat-bot is made up of a menu with five options which are: Checking for panic disorder and agoraphobia, overcoming panic disorder, exploring panic disorder, recording and reviewing my daily history and configurations, as well as the possibility of sending messages such as "help me". With the investigation it was specified that the chatbot represents a positive therapeutic tool with wide access, self-management and interactivity with its patient.

In [12] they propose to change the lifestyles of incorrect nutrition and little physical activity in people, to avoid death and chronic diseases. All this, with the implementation of a chatbot which will guide people with an introductory session, weekly check-ups and goal setting. of behaviors to have a healthy diet and avoid chronic diseases.

In [13] an analysis was developed to reduce the difficulties and obstacles found in the inspection tasks and their ineffective management involved. All this, with the implementation of devices and interfaces that guide and notify in a simple and understandable way about the necessary events to resolve. This research allowed the development of a chatbot that reduces the complexity of building inspections, paperwork problems and optimizes administrative management.

In [14] the development of a discussion support system based on an online discussion moderator agent that can give fluidity to the debates that take place is proposed. Said conversational agent is identified as a chatbot or social bot with the ability to identify the arguments of the participants, structure a facilitating message and publish it to continue the discussion. With this implemented system, it was possible to generate greater fluidity in the organized debates, increase the participation of users and cause more ideas to be found for the topics debated.

In [15] they developed a chatbot, which carries out debates in search of a persuasive technology to promote online on how to implement development goals domestic recycling in young people from Sweden. It was carried out a sustainable in the city of Afghanistan. Two types of debate were chosen for this experiment, polarized and non-polarized. With the proposed experiment, it is concluded that the chatbot influences the ideas of the participants who do not have any knowledge about the topics to be discussed. However, it reinforces the postulate of the participants that you have prior knowledge.

Finally, in [16] the relevance of a new incorporation of learning through the chatbot as a new training comparable to face-to-face learning was investigated. The purpose was to demonstrate comparable efficacy, as well as intrinsic and beneficial attributes for students. The use of chatbot for learning in students showed that the students developed a

greater self-motivation than the group of students who did the learning in a traditional way.

IV. PRINCIPAL CONTRIBUTION

To find the best solution to the problem encountered, had investigated current persuasive technologies. research of [17], persuasive technologies were categorized into twelve different platforms, which are shown in table I.

TABLE I

OVERVIEW OF PERSUASIVE TECHNOLOGY PLATFORMS TO ADDRESS GENERAL CHALLENGES, E.G. MOTIVATION, WORK AND COMMUNICATION, EXTERNAL CONSTRAINTS, ETC.

Technology platform	Application scenario
AR and VR	Enhancing acting and enacting by displaying additional information
Displays, Lights and Sound	Ambient lighting or displays for influencing participant behavior
Games	Influence users to participate, reflect opinions, or to enable learning
Online Collaboration Tools	Virtualize talking, telling and explaining methods, by using e.g. video conferences, collaborative blogs, wikis, etc
(Persuasive) Messages	Structure and guide through conversations with persuasive messages
Physical Tags	Trigger location-based participation
Prototyping Tool kits	Technology-focused prototyping
Robotics	(Humanoid) robot guiding through processes or assisting e.g. observations
Sensors, Analytic and IoT	Simplify e.g. timelines and self observations
Smartphone App	Digitizing various talking, telling and explaining activities by using smartphone apps
Virtual Agent and AI	Virtual agent or AI monitors engagement and persuades passive people
Web and Social Network	Motivate participants by virtualizing parts of the design process using a persuasive social network

A. Selection

Based on the previously presented research of [17]and the study by Christina Bremer [4], which is based on the search for a persuasive technology to promote home recycling in young people in Sweden. A combination of the criteria proposed by Christina Bremer with some of our own research criteria was made, and we selected three technologies that meet the highest percentage of these criteria. The selected technologies are the following:

- Virtual assistant (Chatbot): It is a software program that is normally used to initiate or carry out a dialogue through text in order to resolve recycling issues.
- Serious Games (Garbage Collection Game): It is a gamified system that aims to encourage better recycling behavior
- Internet of things (Intelligent Garbage Cans): It is a tool based on IoT that allows minimizing the resources used in the collection of urban waste.

It is for this reason that our research is based on developing a chatbot that encourages and/or motivates the habit of recycling in users, in addition to increasing knowledge on solid

TABLE II
BENCHMARK OF PERSUASIVE TECHNOLOGIES TO GENERATE RECYCLING HABITS

Feature/Technology	Chatbot	Smart Trash	Serious Game
Easy access to optimal HRB information	X		X
Use of various motivational strategies	X		X
Recognition of differences in recycling schemes	X	X	X
Treat users as equals	X	X	X
Use of an easily accessible technology channel	X	X	X
Low cost of implementation	X		

waste management issues. by providing quality information in real time. For this reason, different art studies were carried out in order to demonstrate that our technological solution was the most suitable.

B. Implementation

For the implementation of our chatbot, the Watson Assistant service from IBM [9] was used, where the entities, intentions and dialogues were created so that it is properly trained and resolves any questions about recycling issues. Likewise, IBM gives you the option of creating an API so that the chatbot can connect to various channels (Facebook, Slack, Twitter, etc). For this project the API was integrated with our mobile application where people can interact with the various modules, for example:

- The chatbot module: Section to answer questions about recycling issues through text or audio.
- The news module: Section that allows us to view current news on recycling and solid waste issues to reinforce the knowledge acquired in the chatbot section.
- The questionnaire module: Section that allows us to test the acquired knowledge of the target audience on recycling issues in the chatbot section.
- The recycling map module: Section that allows us to view the closest places to recycle various materials.

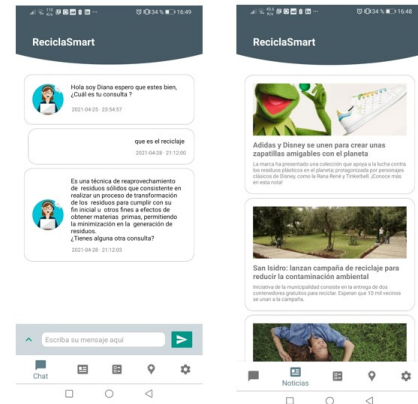


Fig. 1. Chatbot and News

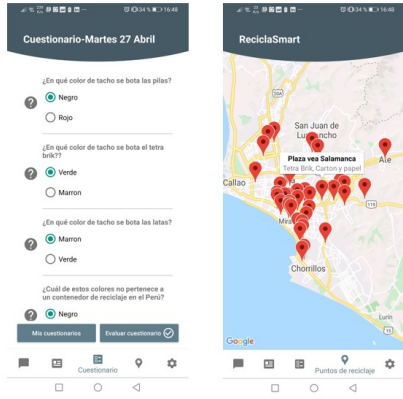


Fig. 2. Questionnaire and Map

For the back-end, the Azure App Service was used to deploy our mobile application and web application. The latter is synchronized with the mobile application, since it allowed us to add, delete, update the information of the news module, questionnaires and recycling map. Finally, the Azure SQL service was acquired to host our data and information schema.

C. Experiment

After developing our technological solution, we moved on to the experimental phase in which we put the chatbot to the test in a real environment. To determine if the proposed system motivates and/or encourages good practices on recycling issues, we requested the participation of 69 university students aged between 18 and 25 who have knowledge on recycling issues and who reside in Metropolitan Lima. For the analysis of these data used statistical techniques descriptive and inferential, which give us a confidence level of 90% and a margin of error of 10%. Before starting the experiment, we held a meeting with the participants where it was explained about the functionalities of our mobile application and where we share a first questionnaire on basic recycling issues in order to know the level of knowledge and its trends regarding to recycling. Second, the mobile application installer was sent so that users have access to our solution and can interact with the ReciclaSmart application for a week. Then, at the end of the validation week, a final questionnaire on recycling issues was shared to validate whether it was possible to increase the users' knowledge and feel more motivated to carry out recycling actions.

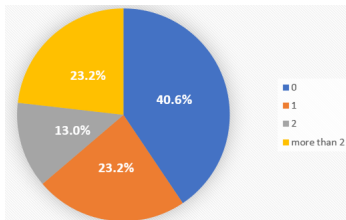


Fig. 3. How many times have you recycled in the last month?

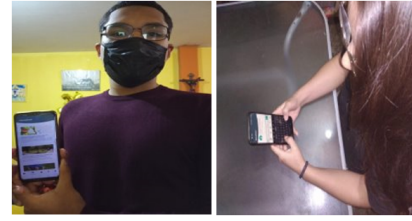


Fig. 4. Participants

V. RESULTS

For the validation, the following indicators were established that allowed us to know the impact of our chatbot on the participants:

- Percentage of users who consider themselves motivated to recycle: As initial information according to [18], 47.3% of university students have solid waste recycling and reuse behaviors. This indicator is useful because it allows us to measure the percentage of university students who feel motivated to carry out recycling activities.
- Average search time on recycling and solid waste issues: As initial information, according to Netbull, the digital marketing agency Netbull [19], a person takes 70 seconds to search for any information on the internet. This indicator is useful, since it allows us to know the time in which the chatbot answered any query about recycling or solid waste management.
- Average of correct answers on basic recycling topics: As initial information, a questionnaire on recycling topics was carried out, where the average grade of the participants was 9.6/20.0. This indicator is important, since it allows us to know the increase in knowledge that university students have after using our technological solution.

At the end of the experiment, we used the data obtained to be able to analyze the established indicators and the results were as follows:

TABLE III
RESULTS AT THE START AND COMPLETION OF THE VALIDATION

	BEGINNING	FINISH
INDICATOR 1	47%	96%
INDICATOR 2	70s	15s
INDICATOR 3	9.6	16.4

Users were asked if our application generated or encouraged them to carry out activities related to solid waste management or recycling, as can be seen in the previous graph. 69 participants mentioned that the application provided information Quality control and in real time generated an intention or motivation to carry out pro-environmental activities related to solid waste management and/or recycling.

Has the information provided and the use of the application helped increase your recycling habits?

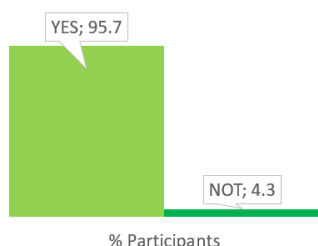


Fig. 5. Graph of average scores obtained in the performed tests

According to the NETBULL Digital Marketing Agency [19], a person takes time to search for information on any topic on the internet around 70s, as can be seen in the table above, the users of our application stated that our solution offers quality and understandable information around 15s, showing an optimization of 78.57% over time search.

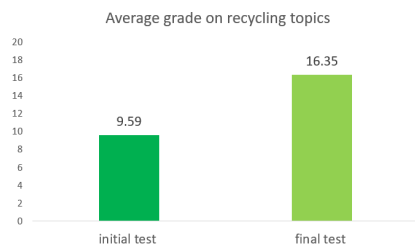


Fig. 6. Graph of average marks obtained in the tests carried out

As can be seen in the graph above, the university students began the validation with a score of 9.59/20 in basic knowledge on recycling and solid waste management. At the end of the validation, the data from the second survey showed that the knowledge of the university participants had increased by 70.49% with the average grade now being 16.35/20.

VI. CONCLUSIONS

In this work, a chatbot was implemented as a persuasive technology to change habits on recycling and solid waste management. With the results obtained, it can be demonstrated that our technological solution encourages and/or motivates the habit of recycling in users, since more than 90% of the participants presented motivation or intention to carry out recycling activities. In addition, our technological solution increases knowledge of the topics investigated in this study by providing quality information in real time, this was evidenced in the difference in the grades of the surveys made at the beginning and at the end. What helped the most in the implementation of this work was the use of IBM Watson services, since it offers us tools aimed at artificial intelligence and chatbot training.

As an opportunity for improvement found in the project, an experiment can be carried out with a larger number of

participants, which would improve the level of confidence in the results we have obtained, the experiment could be carried out in two groups, one controlled and the other controlled. another to act freely, to be able to compare the influence of our solution and the commitment of people, finally, the validation time is key so that people can adapt to our application, which is why Managing a longer time for experimentation would benefit the project.

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